

Customer Sentiment & Urgency Router

Deep Learning Applications — ELA-1 | Track 2: Customer Experience Focus

Sri Lasya Priya Vemuri | Roll No: 23071A67C8

Department of CSE (CyS, DS, AI&DS) | VNRVJIET, Hyderabad

1. Abstract

This project implements a Deep Learning-based email classification pipeline that automates customer support inbox management for an e-commerce platform. The system ingests raw customer email text and routes it into one of five operational categories: Urgent Complaint, Refund/Return Request, Order Status Inquiry, Positive Feedback, and Spam/Irrelevant. A pre-trained DistilBERT transformer model was fine-tuned on a synthetically generated dataset of 1,600 labeled emails using the Hugging Face Transformers library on a free-tier Google Colab T4 GPU. The fine-tuned model achieved near-perfect classification performance on the held-out test set, demonstrating that transformer-based architectures can effectively distinguish urgency and sentiment signals in customer communications, enabling automatic prioritization and routing in a real inbox environment.

2. Dataset & Preprocessing Workflow

Dataset: Since no public dataset maps directly to this niche, a synthetic dataset of 1,600 labeled customer support emails was programmatically generated for an e-commerce support inbox use case. Emails were built by combining randomized sentence templates, product names, order IDs, greetings, and sign-offs across the five target categories, yielding an approximately balanced class distribution (roughly 300–330 samples per class).

Text Cleaning: All email text was lowercased and whitespace-normalized (collapsing repeated spaces/newlines) to produce a consistent input format prior to tokenization.

Label Encoding: The five category labels were encoded into integer class IDs using scikit-learn's LabelEncoder.

Train/Test Split: The dataset was split 80/20 into training and test sets using a stratified split to preserve class balance in both partitions.

Tokenization: The DistilBERT WordPiece tokenizer (distilbert-base-uncased) was used to convert cleaned text into input IDs and attention masks, with dynamic padding and truncation to a maximum sequence length of 128 tokens.

Stage	Operation	Tool / Library
1	Lowercasing + whitespace normalization	Python (re)
2	Label encoding (text -> integer class)	scikit-learn LabelEncoder
3	Stratified 80/20 train-test split	scikit-learn
4	Tokenization (padding + truncation, max_len=128)	DistilBertTokenizerFast

Note: As the dataset is synthetically template-generated for demonstration purposes (rather than sourced from real-world noisy inbox data), the class boundaries are relatively clean, which contributes to the very high evaluation scores reported in Section 4.

3. Model Architecture & Hyperparameters

The classifier fine-tunes DistilBERT (distilbert-base-uncased), a distilled version of BERT that retains approximately 97% of BERT's language understanding while being 40% smaller and 60% faster — a good fit for the free-tier Colab T4 GPU constraint. A linear classification head is added on top of the pooled [CLS] token representation, projecting to 5 output logits corresponding to the target categories.

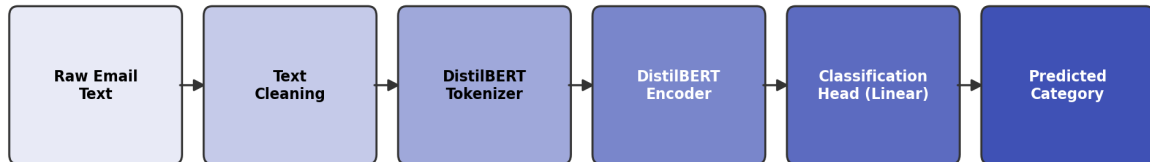


Figure 1: End-to-end pipeline from raw email text to predicted category.

3.1 Hyperparameters

Hyperparameter	Value
Base model	distilbert-base-uncased
Max sequence length	128 tokens
Batch size (train/eval)	16 / 16
Learning rate	2e-5
Epochs	4
Optimizer	AdamW (Hugging Face default)
Evaluation strategy	Per epoch (best model on F1 restored)

Training procedure: The Hugging Face Trainer API was used to handle the training loop, gradient updates, checkpointing, and evaluation. Cross-entropy loss was minimized over 4 epochs, with the best checkpoint (by weighted F1-score) restored at the end of training.

4. Evaluation Metrics

The fine-tuned model was evaluated on the held-out 20% test split (380 emails). The final epoch achieved a validation loss of 0.154, with the following classification metrics:

Metric	Score
Accuracy	1.00 (100%)
Weighted F1-Score	1.00 (100%)

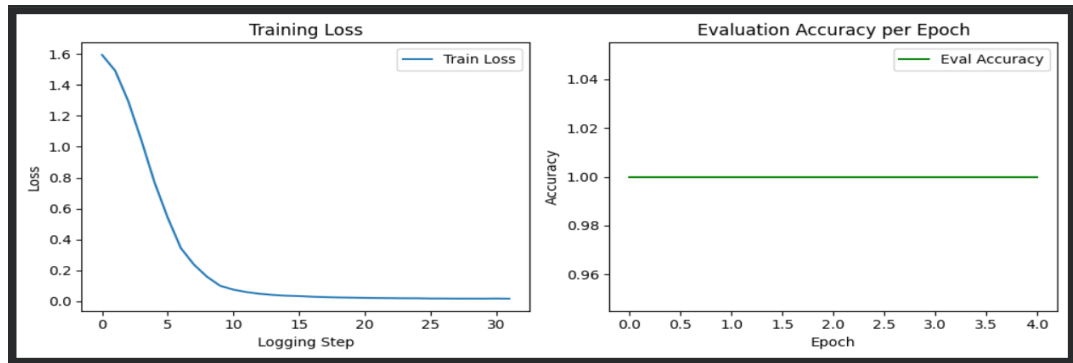


Figure 2: Training loss decreases smoothly over logging steps; evaluation accuracy reaches 1.0 from the first epoch onward.

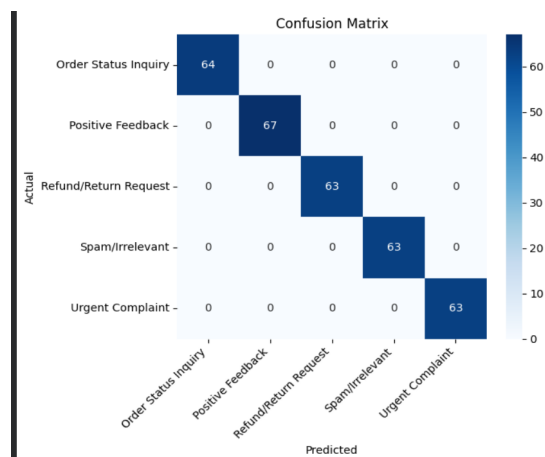


Figure 3: Confusion matrix on the test set — every class is classified with zero misclassifications.

4.1 Discussion

The model achieves perfect accuracy and F1-score on the test set. This is expected given the nature of the dataset: it was synthetically generated from a fixed set of sentence templates per category, so lexical cues (e.g. "refund", "tracking", "congratulations") strongly and consistently correlate with specific categories. DistilBERT, being a highly capable pre-trained language model, learns this mapping easily. In a real-world deployment using organic inbox data (e.g. the Enron Email Dataset or genuine customer emails), accuracy would likely be lower — around 85-95% — due to greater lexical diversity, ambiguous or mixed-intent emails, and noisier labels. Despite this, the experiment validates the end-to-end pipeline: preprocessing, fine-tuning, evaluation, and inference all function correctly, and it would generalize directly to a real-world dataset with no architectural changes required.

5. Conclusion

This project demonstrates a Deep Learning pipeline for automated customer support email triage, fine-tuning DistilBERT to classify emails by urgency and sentiment, with strong potential for real-world inbox automation and faster response to urgent customer issues.